

Periodic ODE Application

System: $\mathbf{x}'(t) = \mathbf{f}(t, \mathbf{x}(t))$

Seek: T -periodic solutions

$$\mathbf{x}(t + T) = \mathbf{x}(t)$$

Construct operator S :

$$S[\mathbf{x}](t) = \mathbf{x}_0 + \int_0^t \mathbf{f}(s, \mathbf{x}(s)) ds$$

Fixed point of S gives periodic solution

Degree Theory Application

Use degree $\deg(I - S, B, 0)$

Key step: Show $\deg(I - S, B, 0) \neq 0$

Then by existence property:

$$\exists \mathbf{x}^* \in B : \mathbf{x}^* = S[\mathbf{x}^*]$$

This $\mathbf{x}^*$ is a periodic solution

Can verify existence without solving explicitly